

# Boundary (*B*) Object Artifact in Dimensional Human Field Theory (DHFT)

DHFT Technical Note

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## Abstract

Boundary (*B*) is a canonical object artifact within Dimensional Human Field Theory (DHFT).

It preserves the invariant identity of Boundary as the structural mediator of load transfer and system differentiation.

Boundary remains non-interpretive and non-substitutable across all admissible representations.

This artifact does not define system-level admissibility or use conditions.

## I. Object Designation

**Canonical Name:** Boundary

**Symbol:** B

**Object Class:** Invariant Variable

**System Location:** Core Mechanical Substrate

Boundary is a primary variable of DHFT.

## II. Canonical Definition

Boundary (*B*) is the structural constraint that mediates load transfer and maintains system differentiation.

Boundary is structural.

Boundary is amoral.

Boundary is non-interpretive.

Boundary is not defined as communication, preference, or negotiation.

## III. Object Function

The function of Boundary is to regulate:

- entry of load
- containment of load
- transfer of load between systems

Boundary participates in system determination only in relation to:

- Load (*L*)
- Capacity (*C*)

## IV. Invariant Properties

Boundary preserves the following invariant properties:

- Directional: governs flow of load
- Mediating: controls transfer conditions
- Structural: independent of interpretation
- Integrity-dependent: may be intact, porous, or compromised
- Determinative: affects transfer and containment outcomes

These properties are invariant.

## **V. Prohibited Substitutions**

Boundary must not be substituted with:

- communication
- boundaries (in social sense)
- limits (in preference sense)
- negotiation
- agreement

No synonymous, approximate, or domain-specific term may replace Boundary within canonical system use.

## **VI. Admissible Operations**

Admissible operations include:

- evaluation of mediation effectiveness
- analysis of permeability
- load transfer analysis
- integrity assessment

## **VII. Non-Admissible Operations**

Non-Admissible operations include:

- interpreting Boundary as communication
- treating boundary failure as misunderstanding
- moralizing boundary conditions
- redefining Boundary through social constructs

Such operations constitute non-admissible representation of Boundary and indicate structural drift.

### VIII. Relational Constraint

Boundary mediates load.

Boundary governs interaction between Load and Capacity.

System state is defined by the relation among L, C, and B.

Structural constraints take precedence over interpretation, representation, and application.

### IX. Representation Constraint

Admissible representation of Boundary must:

- preserve the symbol ( $B$ )
- preserve its mediation function
- preserve non-interpretive use
- preserve relation to L and C

Representations that violate these conditions are non-admissible.

### X. Object Admissibility Condition

A representation is admissible as Boundary ( $B$ ) only if it:

- preserves the canonical symbol ( $B$ )
- preserves the canonical definition
- preserves invariant properties
- does not introduce interpretive or domain-specific substitution
- maintains structural relation to L and C

Violation constitutes non-admissible representation and indicates variable-level structural drift.

## **XI. Completeness Requirement**

Boundary must be treated as a complete structural variable.

Partial extraction of mediation properties does not constitute admissible representation.

## **XII. Measurement Condition**

Boundary must be evaluable in terms of mediation conditions, including:

- transfer
- containment
- permeability

Evaluation must preserve structural relations.

## **XIII. Object Integrity Condition**

Boundary remains the same object across:

- canonical publications
- technical notes
- diagrams
- interfaces
- implementations

Variation in expression does not alter object identity.  
Alteration of mediation function does.

#### **XIV. Verification Reference**

Integrity is preserved through:

- canonical DHFT references
- object definitions
- future verification mechanisms

#### **XV. Forward Compatibility**

This definition operates at the substrate level and remains admissible across future formalizations.

#### **XVI. Canonical Statement**

Boundary ( $B$ ) is a primary invariant object of DHFT.

It mediates load transfer and maintains system differentiation.

It remains fixed, non-interpretive, and non-substitutable across all admissible representations

#### **XVII. System Reference**

The system is specified in:

*Dimensional Human Field Theory (DHFT): Architectural Identification of a Closed Explanatory System*

DOI: 10.5281/zenodo.19075948

All valid system descriptions reduce to Load ( $L$ ), Capacity ( $C$ ), and Boundary ( $B$ ).

No additional variables are introduced at the minimal layer.

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This work defines original structural models and associated terminology within Dimensional Human Field Theory (DHFT).

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